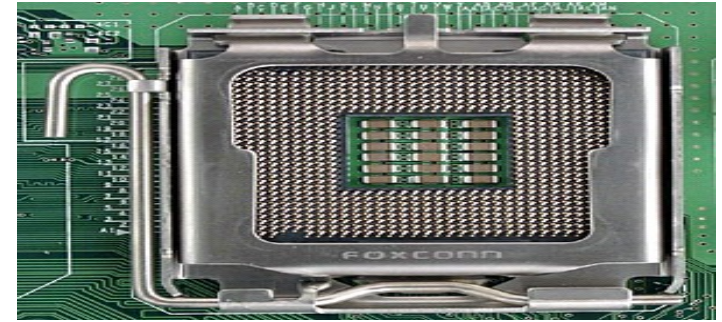
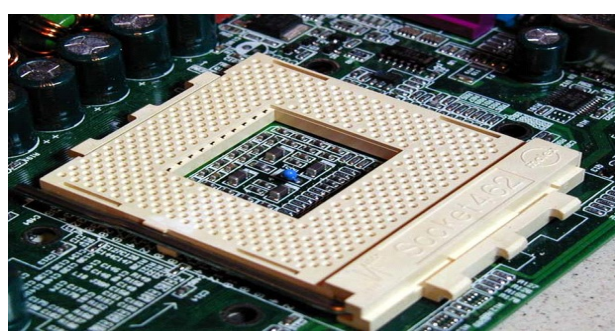




CPU



Demo

[Video\🖥️ - See How a CPU Works.mp4](#)

CPU

- ☑ CPU, also called **microprocessor** or simply processor, is the **brain of PC**.
- ☑ It performs the system's data processing activity like calculation, comparison, etc.
- ☑ It is the **most expensive** single component in the system, costing four or more times than the motherboard it plugs into.
- ☑ The **first CPU** was created in **1971 by Intel** called **4004**.

CPU

- ☑ **Brain of PC:** it directly or indirectly controls the functions and activities of your computer.
- ☑ It transmits information between devices and tells the devices what to do.
- ☑ This is done by sending **control** information.
- ☑ Major producers of CPU are:
 - ☑ **Intel, AMD, Cyrix** and **Sparc**

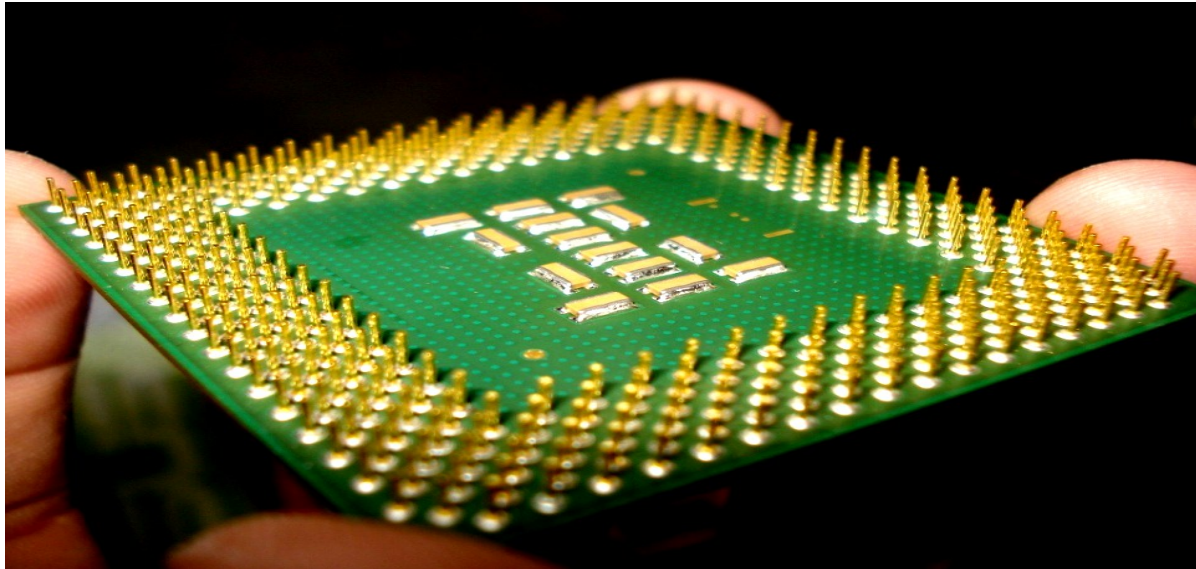
Front Side Bus

- ☑ Front side bus or **system bus** is a data bus that carries all information between CPU and all other devices within the system: RAM, AGP Video Card, PCI Expansion Cards, hard disk, etc.
- ☑ It is also called **data bus**.
- ☑ **All devices are connected to the bus.**
- ☑ Because of this, any information placed on the bus is available to all devices connected to the computer.

Processor Sockets and Slots

- ☑ **Socket:** CPU socket is the connector that link CPU to motherboard particularly those compatible with **Intel x86 architecture**.
- ☑ Most CPU today are built around **PGA(Pin Grid Array)** architecture where the **pins** in the underside of CPU are inserted into the **socket**.
- ☑ In contrast to this, several current and upcoming CPUs use **LGA(Land Grid Array)** where the **pins** are on the socket and they come in contact with the processor.

Processor Sockets and Slots...



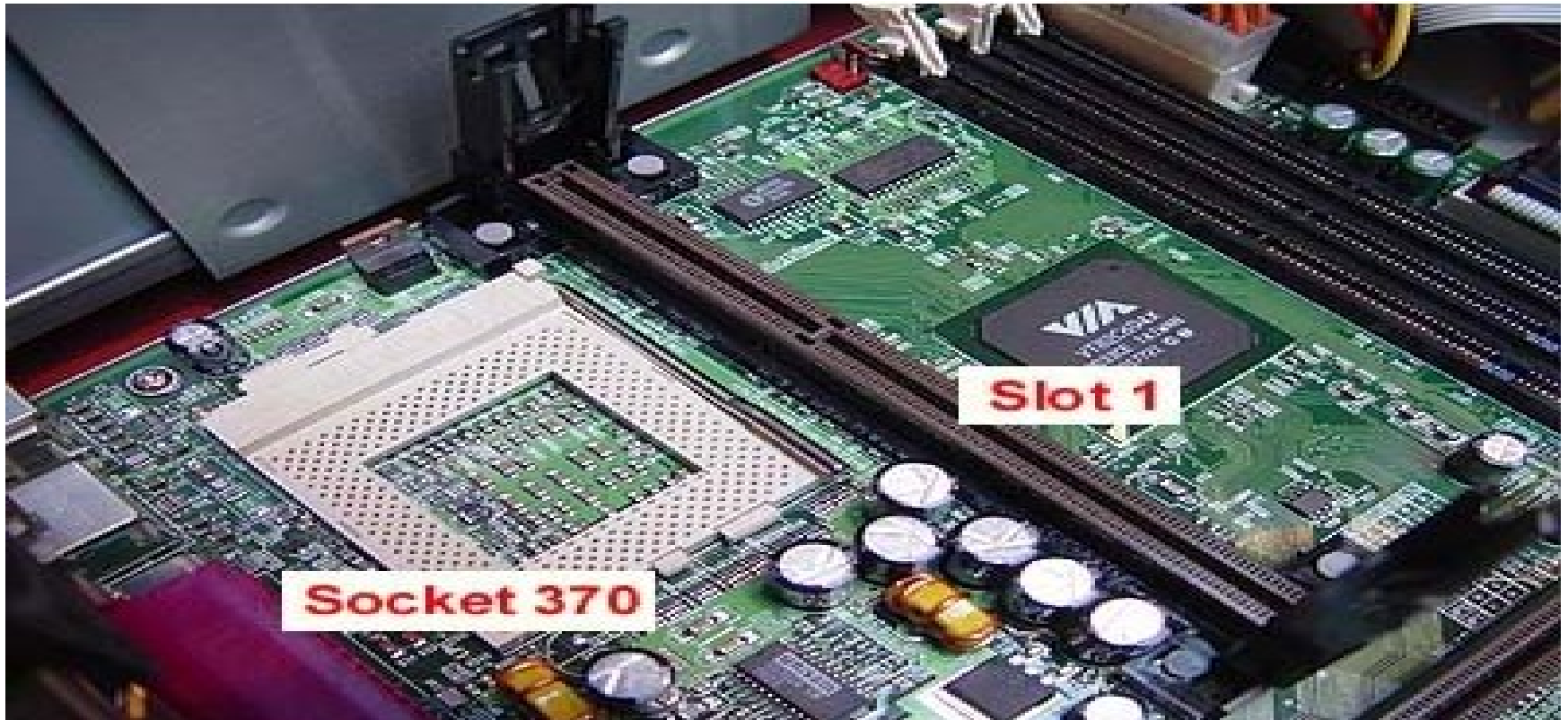
☑ Fig CPU pins that fit to sockets

Processor Sockets and Slots...

- ☑ **Slot:** in this case, processors are **cartridge shaped** and fix into slot that looks like expansion slot.
- ☑ The CPU looks **like card**.



Processor Sockets and Slots...



Both slot and socket are used to connect CPU to motherboard.

Processor Sockets and Slots...

Socket number	Pin number	Pin layout	Supported processors
Socket 1	169	PGA	486
Socket 2	238	PGA	486
Socket 3	237	PGA	486
Socket 4	273	PGA	Pentium
Socket 5	320	SPGA	Pentium, AMD K5
Socket 6	235	PGA	486
Socket 7	321	SPGA	Pentium, AMD K5/K6, cyrix MI/II
Socket 8	387	SPGA	Pentium Pro
Socket 370	370	SPGA	Pentium III, Celeron
Slot A	242	Slot	AMD Athlon
Socket A	462	PGA	AMD Athlon, AMD Duron
Slot 1	242	Slot	PII/PIII, Celeron
Slot 2	330	Slot	PII/PIII Xeon
Socket 423	423	SPGA	Pentium IV
Socket 478	478	PGA	P IV, Celeron
Socket T (LGA 775)	775	LGA	P IV, Celeron D

Socket T: uses Land Grid Array and is the latest socket type.

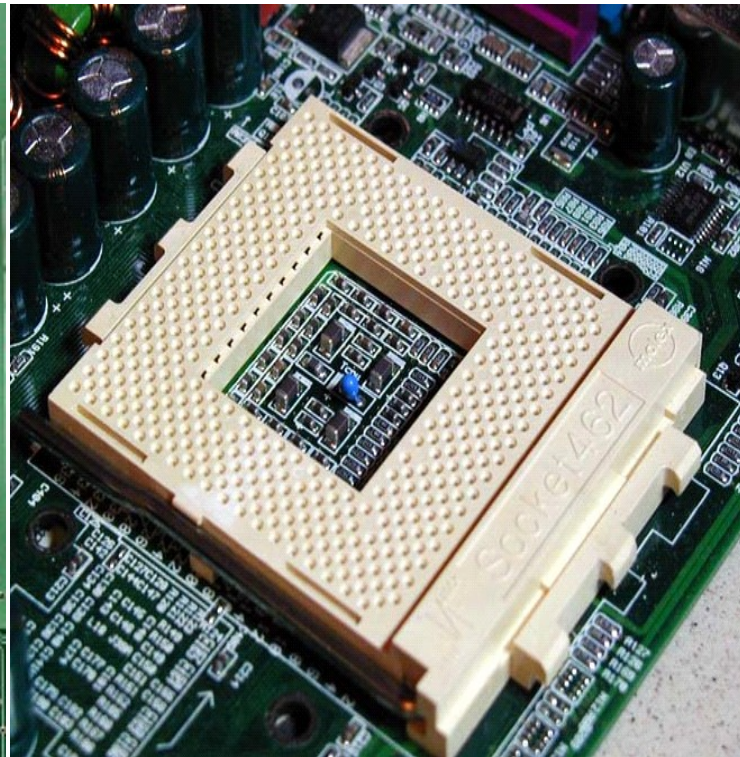
PGA: Pin Grid Array

SPGA: Staggered Pin Grid Array

Processor Sockets and Slots...



Fig CPU
sockets

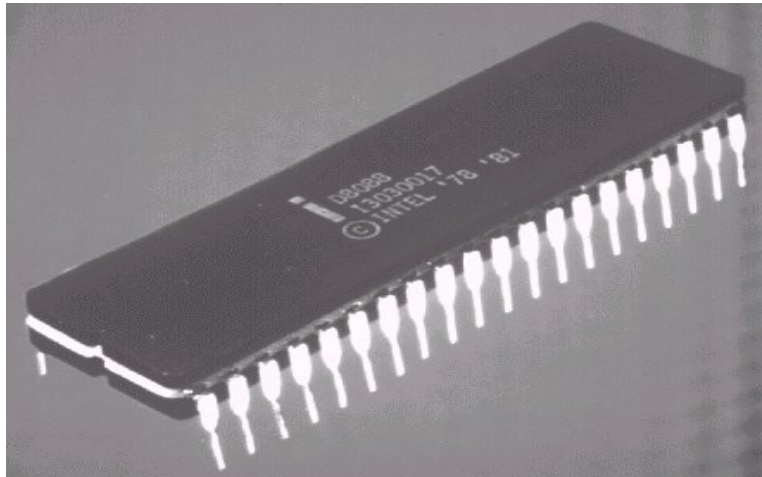


(LGA and SPGA) Respectively

CPU Socket Types

1) **Dual Inline Pin Package (DIPP):**

- ☑ It has a set of pins that protrude in **two** sides of the CPU. Used in 8086, 8088, 80286 CPUs.



CPU Socket Types...

2) Pin Grid Array (PGA):

- ☑ The pins form straight line.



CPU Socket Types...

3) Staggered Pin Grid Array (SPGA):

- ☑ The pins form **diagonal** line.



CPU Socket Types...

4) Single Edge Contact (SEC):

- ☑ no pins and it uses slot.



www.cpu-world.com

CPU Cooling

- ☑ Specialized cooling system became serious starting from **80486**.
 - ☑ Earlier chips ran at a low speed and contained relatively few transistors.
 - ☑ Because of this they needed no specialized cooling.
 - ☑ But increase in CPU speed increased the heat generated necessitating cooling system.
- ☑ CPU cooling uses heat sinks.
 - ☑ Heat sinks conduct heat from CPU to heat sink and then radiate it to air.
- ☑ Good cooling depends on the transfer of heat between the CPU and heat sink.
- ☑ The heat sink and CPU should contact each other to the **maximum surface area** possible. This allows heat to flow easily.

CPU Cooling...

There are two cooling mechanisms:

1. Passive Heat sink:

- ☑ Have **no moving parts**.
- ☑ They are made up of **aluminum**.
- ☑ They cool the CPU by using **thermal conduction and radiation**.
- ☑ The heat sink draws heat from the CPU and air flowing through the heat sink cools the heat sink itself.
- ☑ The heat sink is clipped or glued to the processor.
- ☑ **Thermal compound/grease** is used to glue heat sink and CPU together so that they contact each other fully and there is no gap in between.

CPU Cooling...

2. Active Heat Sink

- ☑ Active heat sink **adds a small fan** that blows directly onto the heat sink metal to ensure direct air flow.
- ☑ An active heat sink **cools better** than passive heat sink by **forcing air flow**. Unfortunately, the fans have short life span and they are the **first thing to fail** in most PCs.

CPU Cooling...

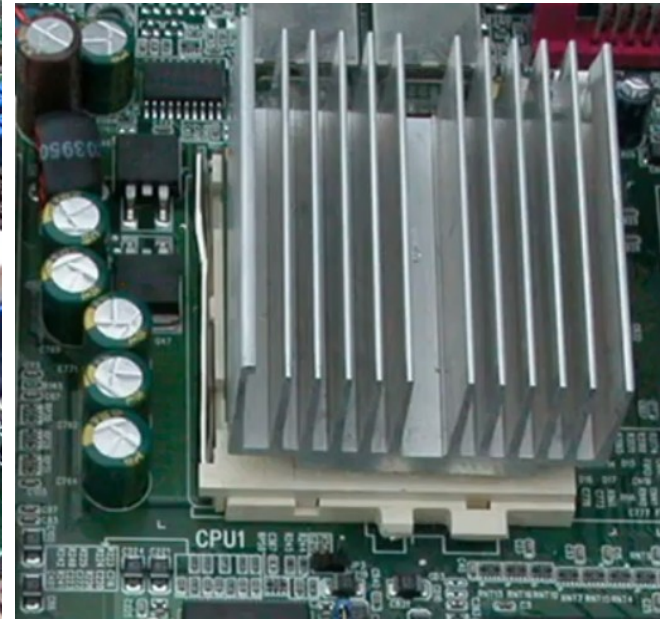
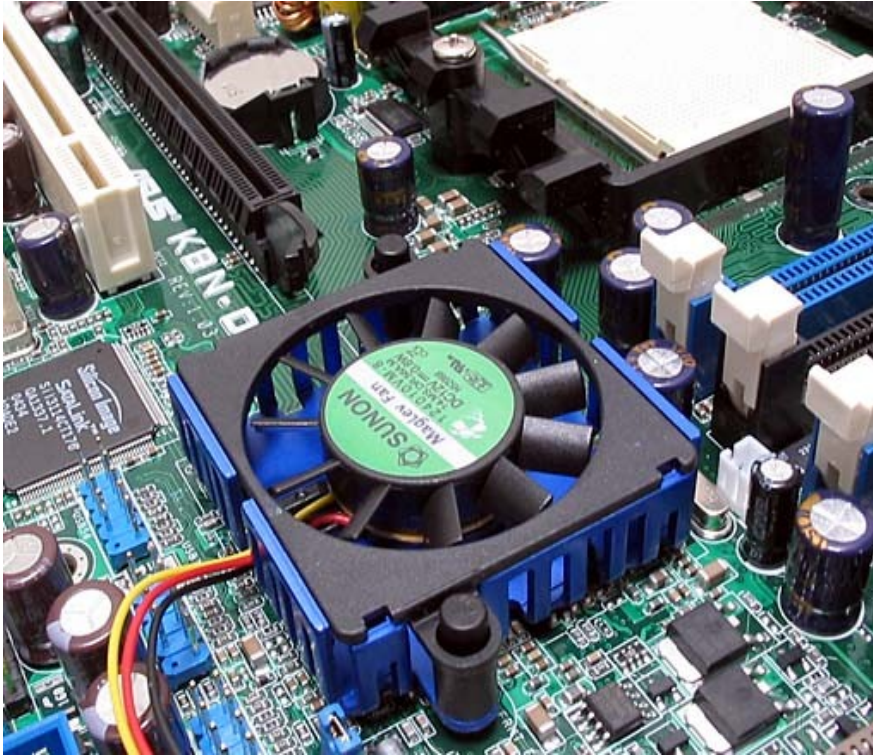


Fig CPU fan and heat sink

CPU Cooling...

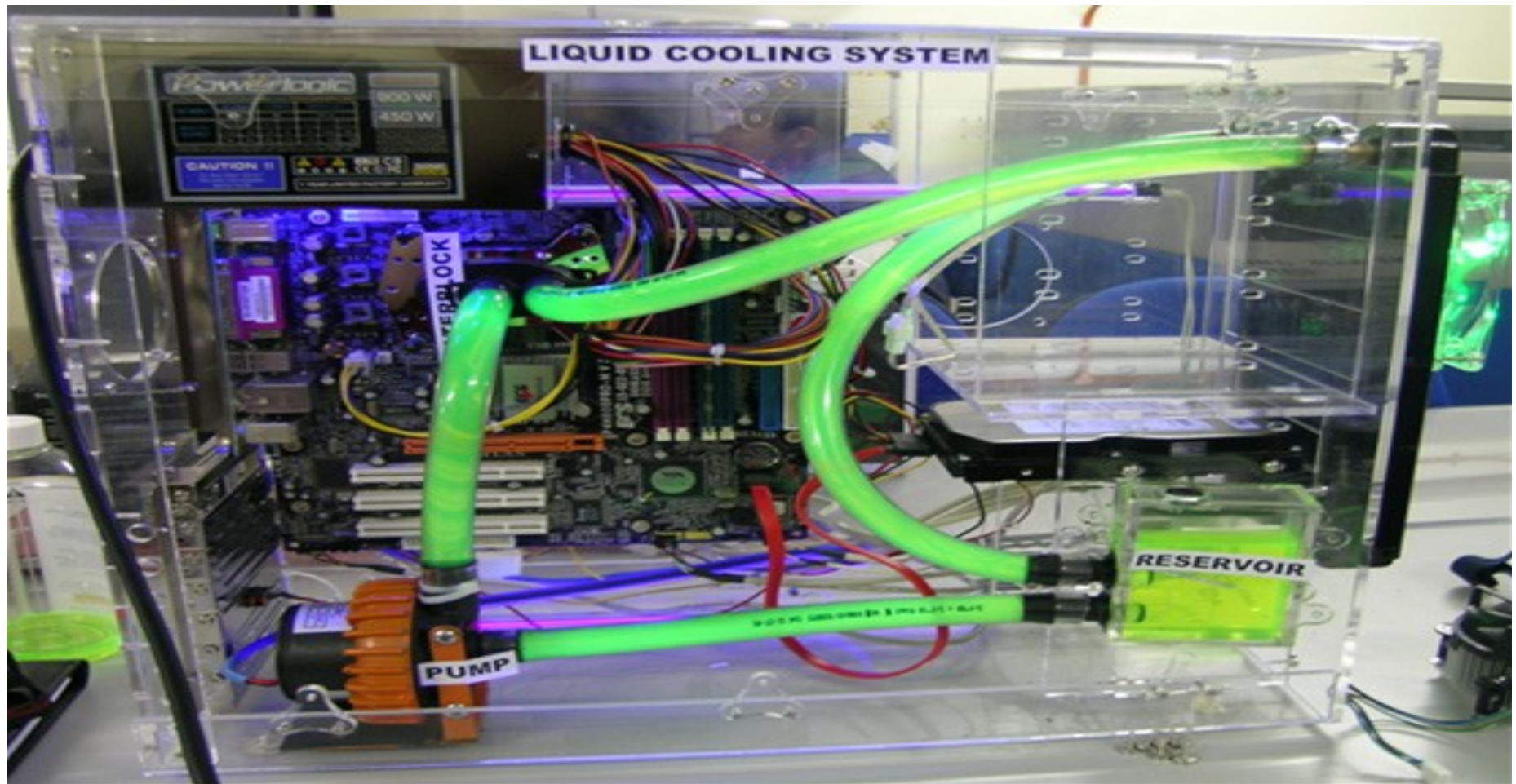
Liquid Cooling

- ☑ On high end systems, liquid cooling is used
- ☑ Common in **Mainframe** computers
- ☑ It has reservoir, pump, and fan

Liquid cooling has three main parts:

- ☑ **Reservoir** where the liquid is stored
- ☑ **Pump** that pushes the liquid through pump to the CPU or other chipsets that should be cooled
- ☑ **Radiator** (fan) that cools the liquid

CPU Cooling...



CPU Cooling...

Overheated CPU **causes**:

- ☑ System crash
- ☑ Random reboot
- ☑ Memory errors
- ☑ Disk problems
- ☑ Application errors: Like Looping

Replacing and Upgrading CPU

Upgrade:

- ☑ installing another device which is **better than** the previous one in terms of performance.
- ☑ Poor upgrade may lead to total failure.

Replace:

- ☑ substitute with equivalent device in case the **previous one failed**.
- ☑ One thing to consider when we replace/upgrade CPU is the **CPU socket** on your motherboard.

Three most common sockets are:

- ☑ Low Insertion Force(LIF)
- ☑ Zero Insertion Force(ZIF)
- ☑ Single Edge Contact(SEC)

Replacing and Upgrading CPU...

LIF

- ☑ Removing an old CPU requires force. There are **special tools** that are designed for this.
- ☑ Use these tools:
 - ☑ **Flat head screw driver**
 - ☑ **Chip puller**

ZIF

- ☑ In ZIF, no force needed to remove or insert CPU into its socket.
- ☑ It is the **most popular socket** for PCs.
- ☑ It is easy to remove and insert CPU.
- ☑ It is introduced in early 1990s as a means to provide **user-friendly** CPU upgrade.

Replacing and Upgrading CPU...

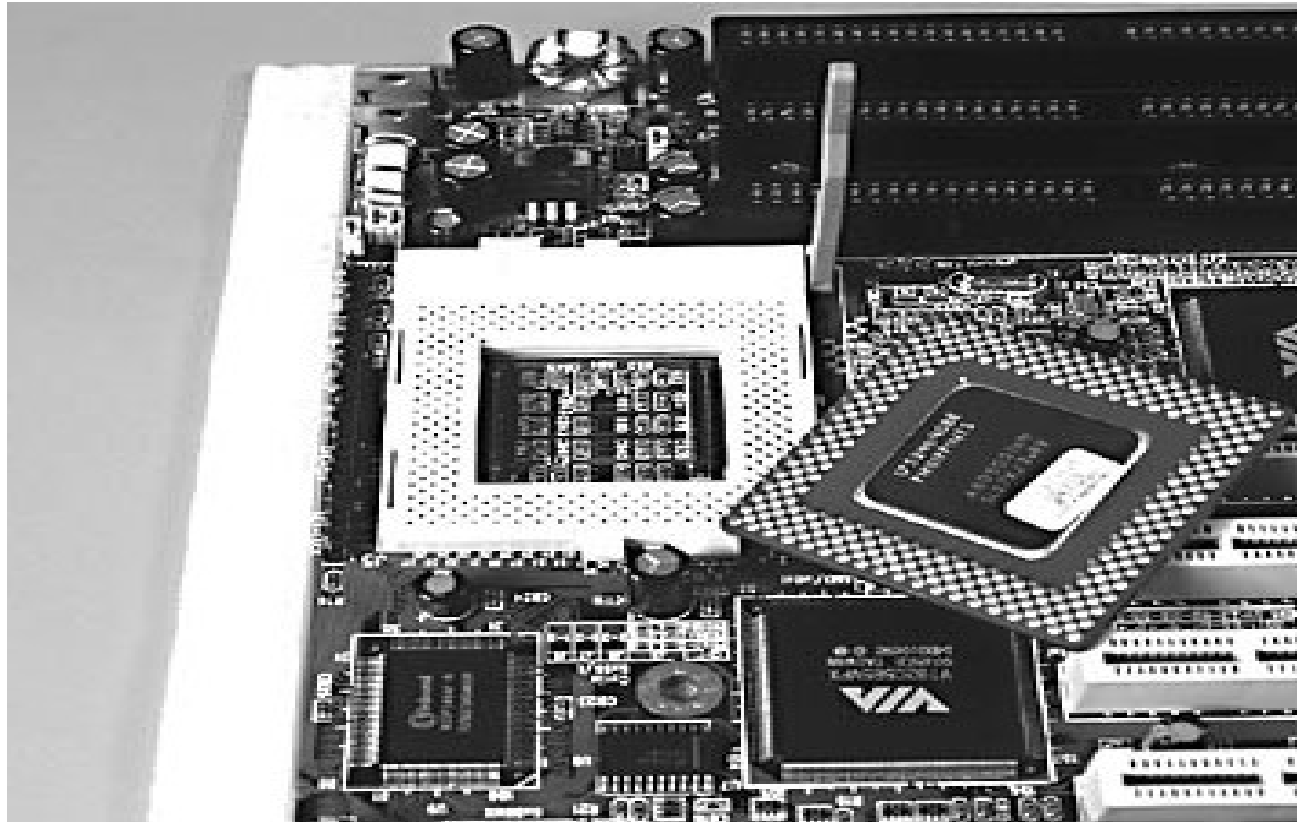


Fig ZIF socket and CPU

Replacing and Upgrading CPU...

- ☑ To **maximize performance**, you always **upgrade** to the fastest processor your particular board will support.
 - ☑ This is determined by the type of socket on your motherboard.
 - ☑ For example, if your motherboard has slot 1, you can't upgrade to Pentium IV which uses socket type connector instead of slot.
- ☑ Therefore, when you **buy** a new CPU for upgrade, select the type of CPU that **matches** with the socket on your motherboard.

Replacing and Upgrading CPU...

Upgrade Advice:

- ☑ Be careful when you handle CPU or any exposed IC. Static discharge could damage the chip.
- ☑ Take care not to bend any pins
- ☑ If you encounter any resistance, stop and figure out what is wrong. Do not use any force to insert the CPU.

Troubleshooting CPU Problems

- ☑ If the CPU does not work, the system gives **beep sounds** when turned on.
- ☑ The number of beeps varies from BIOS to BIOS.
 - ☑ On AMI BIOS, **five beeps indicate CPU problem.**
 - ☑ Visit the **internet** for list of such errors.

CPU problems and trouble shootings:

- ☑ **CPU problem Possible causes:**
 - ☑ Overheating.
 - ☑ Static discharge.
 - ☑ Bent or Broken pins.

General symptom of CPU

- ☑ The system fails to Boot (start).
- ☑ **Black Screen.**
- ☑ The system boots, but the operating system (windows) fails to load.
- ☑ The system locks-up or dies after several minutes of operations.
- ☑ The system says "**hardware monitor error** "

Problem

☑ Both the CPU and power supply fan work Properly but the system shows a **black screen**.

☑ **Solution**

☑ The CPU might be damaged.

☑ Use the replacement method.

☑ Check Functionality of CPU.

☑ Check the damaged/crushed/bent CPU legs.

☑ use the replacement method.

☑ Check the CPU Model with the Motherboard.

Installing CPU:

- ☑ Turn off the PC and unplug the power cable
- ☑ disconnect external devices like mouse, keyboard, etc.
- ☑ follow appropriate ESD procedure
- ☑ remove the cover/case
- ☑ install CPU in its socket/slot
- ☑ install heat sink and fan (don't forget fan power)

Thank You

End of the chapter